👤 USER
IMAGE MASTER
Image Request
Image / Asset Check
MISSING
FOUND
IMAGE SOURCES / PROVIDERS
Image Provider
Original / Source Image
Image Resource
Existing File
Return Stored Resource
Cache Hit
API NOT CALLED
Cache Miss
External API CALLED
AUDIO FOUND
TTS MASTER
TTS Request
article_id + language + voice_id
Audio Cache Check
AUDIO MISSING
TTS PROVIDERS
Google Cloud TTS
Sarvam AI
Generated MP3
Return Cached Resource
Normal Operation
TRANSLATION MASTER
Translation Request
Article Content Hash
Translation Cache
CACHE MISS
CACHE HIT
Translated Text
CLOUDFLARE R2 — OBJECT STORAGE
WebP Images
MP3 Audio
Regeneratable Generated Assets
Return Translation
Return MP3
TURSO — OPTIONAL TRANSLATION CACHE
Cached Translations
FOUND
NODE.JS BACKEND / API PROXY
Storage Monitor
API Router
Cache Service
CACHE HIT
Speech / TTS Service
Translation Service
Media Service
CACHE MISS
Asset Registry Service
ASSET FOUND
ASSET NOT FOUND
Storage Service
Generation Lock
Provider Usage Logger
Analytics Service
Automatic Cleanup Service
TRANSLATION PROVIDERS
Google Translation API
GTX
Clients5
MyMemory
STORAGE MONITORING
Storage Usage
Threshold Check
Normal
>= 85%
>= 85%
Admin Warning
USAGE-BASED AUTOMATIC CLEANUP
Find Low-Usage Assets
Usage Score
Protection Rules
Delete Regeneratable Cache
Target ~70–75%
ADMIN / DEVELOPER DASHBOARD
PublicSpark Admin Dashboard
Storage Statistics
Cache Hit / Miss
API Calls
Asset Saved / Missing
Article Usage Statistics
Cleanup Statistics
Low Usage Articles
NEVER AUTO-DELETE
Protected Important Asset
Original Article
Published Canonical Data
SUPABASE POSTGRESQL — MASTER DATABASE
provider_usage_log
Articles
Article Metadata
Asset Registry
view_count / like_count
article_stats_daily
Universal article_id UUID
PUBLICSPARK FRONTEND
React / PublicSpark UI
Language / Translation UI
Article Page
Audio Player
Browser Cache
ANALYTICS
Article Open
last_viewed_at
Visitor / Session Validation
Atomic view_count +1
Atomic like_count +1
Daily Aggregate
VIDEO
video_id / video_url
YouTube